DefectLens: Interpretable Ensemble Convolutional Networks for Artwork Defect Detection

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***Abstract*—The physical condition of artworks degrades continuously due to surface cracking, chromatic fading, and mechanical abrasion. Existing deep-learning solutions rely predominantly on single-backbone architectures trained under controlled photographic conditions, which generalise poorly across the stylistic and institutional diversity of real museum collections. Moreover, these systems return binary labels without spatial justification, leaving conservation professionals no mechanism to validate automated decisions. We introduce *DefectoScope*, an interpretable ensemble that pairs an efficiency-oriented MobileNetV3Large backbone f_1 with a multi-scale InceptionV3 backbone f_2 under the parameter-free averaging rule $\hat{p}(\mathbf{x}) = \frac{1}{2}[f_1(\mathbf{x}) + f_2(\mathbf{x})]$, augmented by three-orientation test-time augmentation (TTA). Gradient-weighted Class Activation Mapping (Grad-CAM) produces per-pixel saliency maps exposed through a Flask web interface, enabling conservators to inspect the spatial evidence underlying each prediction. We derive a variance-reduction guarantee for the averaging estimator, prove the balanced-risk equivalence of inverse-frequency class weighting, and establish a closed-form expression for the Bayes-optimal decision threshold. Evaluated on the *Art Images: Clear and Distorted* Kaggle corpus—a $\approx$100 GB heterogeneous collection spanning Impressionism, Baroque, Realism, and Renaissance works—DefectoScope achieves an F_1-score of 0.9893 (precision 0.9901, recall 0.9885) on a held-out 20% validation partition. Qualitative Grad-CAM analysis confirms attendance to visually plausible defect regions in the majority of true-positive cases. We discuss three open methodological limitations transparently: annotation uncertainty, the absence of single-model ablation baselines, and uncalibrated confidence scores.**

***Index Terms*—Artwork defect detection, ensemble learning, MobileNetV3Large, InceptionV3, Grad-CAM, transfer learning, binary image classification, cultural heritage, explainable AI.**

I. Introduction

Cultural heritage institutions—museums, galleries, and archival collections—bear custodial responsibility for artworks whose physical substrates degrade under continuous environmental stress. Surface cracking, chromatic fading, waterlogging stains, and mechanical abrasion collectively threaten the long-term integrity of paintings, sculptures, and mixed-media objects. Degradation is driven by humidity fluctuation, thermal cycling, biological growth, and mechanical handling, all of which alter the microstructure of paint layers at rates that are difficult to monitor without systematic imaging programmes [2], [3].

For large collections, manual condition inspection is operationally intractable. A national gallery holding tens of thousands of objects requires person-years of expert effort for systematic photographic survey, and smaller regional institutions often lack specialist imaging equipment entirely [4], [11]. The economic and cultural stakes are high: undetected surface damage that progresses beyond a critical threshold can render an artwork irreparable.

A. *From Hand-Crafted Features to Deep Ensembles*

Early automated defect identification relied on hand-crafted descriptors. Gabor filter banks capture oriented texture statistics at multiple spatial scales and frequencies, while Local Binary Patterns (LBP) encode local contrast in a rotation-invariant histogram [3], [4]. Although computationally inexpensive, these descriptors lack the representational capacity to discriminate genuine surface damage from stylistic texture variation, natural craquelure, impasto brushwork, or photographic artefacts.

Transfer learning from large-scale datasets such as ImageNet [12] enabled convolutional neural networks (CNNs) to generalise beyond narrow texture statistics [4], [5], and deep convolutional architectures have since demonstrated strong performance on related visual inspection tasks in manufacturing and civil infrastructure monitoring. However, these methods are predominantly evaluated on single-style datasets under controlled photographic conditions, and their performance on the heterogeneous, multi-institution imagery of real museum collections has not been rigorously characterised.

A further limitation is the single-backbone paradigm: a model built on one architecture captures one type of inductive bias. MobileNet-family models [8] emphasise channel-wise efficiency through depthwise separable convolutions, producing fine-grained local feature maps at low computational cost. Inception-family models [9] capture multi-scale spatial context through parallel convolutional branches within each module. The complementarity of these biases motivates their combination within an ensemble [6], [7].

B. Interpretability as a First-Class Requirement

Most existing defect detection systems treat predictions as black boxes, returning a binary label $\hat{y} \in \{0, 1\}$ without indicating which spatial region of the artwork drove the decision. In a high-stakes conservation domain this is operationally unacceptable: a professional cannot act on a prediction they cannot inspect or contest [2]. Gradient-weighted Class Activation Mapping (Grad-CAM) [10] addresses this by computing the gradient of the class score with respect to the final convolutional feature map, producing a heatmap that attributes each decision to specific spatial locations without requiring architectural modification.

C. Summary of Contributions

Building on the above motivation, this paper makes the following four contributions:

1) **DefectoScope** (Section IV & VIII): an ensemble pairing MobileNetV3Large and InceptionV3 within a parameter-free averaging rule with three-orientation TTA, together with a Flask interface exposing Grad-CAM saliency maps to conservators.
2) **Theoretical foundations** (Section V): closed-form variance-reduction guarantee for the averaging estimator (Proposition 1), a variance bound for TTA (Lemma 1), the Bayes-optimal decision threshold under asymmetric misclassification costs (Eq. 14), and proof of balanced-risk equivalence for inverse-frequency weighting (Proposition 2).
3) **Grad-CAM formalisation** (Section VII): complete derivation of the Grad-CAM saliency map (Eqs. 23–24) and an architecture-agnostic layer-targeting algorithm (Algorithm 1).
4) **Empirical evaluation** (Section IX): $\mathrm{F}_1 = 0.9893$ on the $\approx$100 GB *Art Images: Clear and Distorted* corpus [1], with a rigorous discussion of methodological limitations that constrain interpretation of this result.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset and preprocessing. Section IV details the model architecture. Section V establishes theoretical properties of the ensemble and TTA estimators. Section VI specifies training and inference design. Section VII formalises Grad-CAM. Section VIII describes the full system architecture. Section IX presents results and comparative analysis. Section X performs error analysis. Section XI offers a broader discussion. Section XII concludes.

II. Related Work

A. Defect Detection in Artworks and Heritage Objects

The earliest CNN-based approaches fine-tuned VGG-16 on single-style datasets (e.g., historical frescoes) and substantially outperformed hand-crafted feature extractors [4]. Hybrid CNN–SVM classifiers [6] combine the representational power of learned features with the maximum-margin inductive bias of support vector machines, achieving competitive recall at the cost of precision under class imbalance. Attention-enhanced architectures [2] provide spatially accurate localisation by learning to attend to defect regions during training. Rothman and Nakamura [3] couple ResNet-50 with post-hoc explainability tools, demonstrating that interpretable predictions increase practitioner trust. Bai et al. [5] survey challenges and advances in machine learning for material defect identification across industrial and art domains. Fiorucci et al. [11] survey machine learning for cultural heritage broadly, identifying ensemble methods as among the most reliable strategies for heterogeneous, limited-labelled datasets.

To our knowledge, no prior work has (i) combined MobileNetV3Large and InceptionV3 within a single TTA-enhanced averaging ensemble for artwork defect detection, (ii) provided theoretical variance-reduction analysis for the resulting estimator, or (iii) integrated Grad-CAM saliency into a conservation-oriented web interface for this application domain.

B. Ensemble Methods in Visual Classification

Rokach [7] establishes the theoretical foundations of ensemble learning: combining classifiers whose errors are imperfectly correlated reduces expected error below the average constituent error. Architectural diversity—using networks with different inductive biases—is a practical proxy for statistical independence. We formalise this result specifically for the two-model arithmetic mean in Section V.

C. Grad-CAM and Explainable AI

Selvaraju et al. [10] introduce Grad-CAM as a class-discriminative localisation technique requiring no architectural modification. Unlike attention maps that must be trained end-to-end, Grad-CAM is a post-hoc technique applicable to any pretrained model, making it particularly suitable for transfer-learning pipelines where the backbone was not designed with interpretability in mind.

III. Dataset and Preprocessing

A. Art Images: Clear and Distorted

Our experiments are based on the publicly available *Art Images: Clear and Distorted* dataset [1], a heterogeneous collection of approximately 100 GB of photographic images partitioned into two top-level classes:

- **Clear** ($y = 0$): intact artworks with no visible surface degradation.
- **Distorted** ($y = 1$): artworks exhibiting one or more of: surface cracks, chromatic fading, waterlogging stains, and mechanical abrasion.

The collection spans stylistically divergent movements—Impressionism, Baroque, Realism, and early Renaissance—under highly varied institutional photographic conditions. Using an 80/20 training–validation split (seed 123), the pipeline yields $N_{\mathrm{tr}} \approx 80{,}000$ training samples and $N_{\mathrm{val}} \approx 20{,}000$ validation samples.

Remark 1 (Annotation Caveat): Defect labels were assigned based on photographic appearance rather than by a certified conservation professional. Consequently, some *distorted* samples may contain photographic artefacts (JPEG DCT blocking, motion blur, specular glare) that superficially resemble genuine surface damage. We treat this as an irreducible source of label noise ε with unknown magnitude and do not attempt post-hoc correction.

B. Normalisation Pipeline

Samples are loaded via TensorFlow's `image_dataset_from_directory` API with target spatial resolution 224×224 pixels and batch size $B = 32$. Each raw pixel value $x_{\text{raw}} \in [0, 255]$ is mapped to $[-1, 1]$ by the affine transformation:

$$x = \frac{x_{\text{raw}}}{127.5} - 1.0, \qquad x \in [-1, 1]. \tag{1}$$

This follows the MobileNetV3 preprocessing convention. Because both constituent models are fine-tuned end-to-end, any normalisation offset induced by applying (1) to InceptionV3 (which conventionally expects $[-1, 1]$ input) is absorbed by the learned affine parameters of the first convolutional layer during fine-tuning.

C. Class-Weight Computation

Let $\mathcal{C} = \{0, 1\}$ and let $n_c = |\{i : y_i = c\}|$ be the number of training samples in class c. The inverse-frequency class weight for class c is:

$$w_c = \frac{N_{\text{tr}}}{|\mathcal{C}|\, n_c} = \frac{N_{\text{tr}}}{2\, n_c}, \qquad c \in \{0, 1\}. \tag{2}$$

These weights satisfy $\sum_{c \in \mathcal{C}} (n_c / N_{\text{tr}}) \cdot w_c = 1$, i.e., the weighted empirical risk equals the unweighted empirical risk on a perfectly balanced dataset, as proven in Proposition 2.

D. Augmentation Strategy

Let $\mathcal{T} = \{T_0, T_1, T_2\}$ be the set of label-preserving geometric transformations:

$$T_0(\mathbf{x}) = \mathbf{x}, \quad T_1(\mathbf{x}) = \text{FlipH}(\mathbf{x}), \quad T_2(\mathbf{x}) = \text{Rot}_{90}(\mathbf{x}). \tag{3}$$

Photometric transforms (brightness jitter, colour contrast, random crops) are deliberately excluded. The rationale is that drastic luminance changes applied to images of gradual chromatic fading can render the defect visually invisible, producing training examples whose augmented appearance contradicts their label. Geometric transforms preserve local texture statistics and therefore do not alter the perceptual severity of crack or abrasion defects.

IV. Model Architecture

A. Backbone Selection and Complementarity

DefectoScope employs two ImageNet-pretrained backbones selected for their architectural complementarity.

1) MobileNetV3Large (f_1): Built on inverted residual blocks with hard-swish activations and squeeze-and-excitation (SE) channel attention [8]. The depthwise separable convolution at its core factorises a standard $k \times k$ convolution with C_{in} input and C_{out} output channels into a depthwise spatial step followed by a pointwise 1×1 step, reducing the multiply-accumulate cost from

$$\mathcal{O}(H\, W\, C_{\text{in}}\, C_{\text{out}}\, k^2) \ \text{ to } \ \mathcal{O}\big(H\, W\, C_{\text{in}}\big(k^2 + C_{\text{out}}\big)\big)\,,$$

a reduction factor of approximately k^2 for $C_{\text{out}} \gg 1$. The backbone produces feature maps $\mathbf{A}_1 \in \mathbb{R}^{7 \times 7 \times 960}$ at approximately 0.22 GFLOPs per image.

2) InceptionV3 (f_2): Employs factorised convolutions ($n \times n$ decomposed into $1 \times n$ and $n \times 1$ convolutions) and parallel branches with kernel sizes 1×1, 3×3, and 5×5 operating simultaneously on the same feature map [9], capturing multi-scale spatial context within each module. The final convolutional output is $\mathbf{A}_2 \in \mathbb{R}^{5 \times 5 \times 2048}$ at approximately 5.5 GFLOPs per image.

The two backbones exhibit orthogonal inductive biases: f_1 is oriented toward channel-wise efficiency and local fine-grained features, while f_2 captures broad multi-resolution representations. This diversity is the primary mechanism for reducing inter-model error correlation ρ, thereby maximising the variance reduction established in Section V.

B. Classification Head

On top of each backbone $i \in \{1, 2\}$ with final feature tensor $\mathbf{A}_i \in \mathbb{R}^{H_i \times W_i \times C_i}$ we attach an identical sparse classification head. Global average pooling (GAP) reduces the spatial dimensions:

$$\mathbf{g}_i = \text{GAP}(\mathbf{A}_i) = \frac{1}{H_i W_i} \sum_{h=1}^{H_i} \sum_{w=1}^{W_i} \mathbf{A}_i[h, w, :] \ \in \mathbb{R}^{C_i}. \tag{4}$$

A single sigmoid unit then produces the scalar probability:

$$f_i(\mathbf{x}) = \sigma\big(\mathbf{v}_i^\top \mathbf{g}_i + b_i\big) \in (0, 1), \tag{5}$$

where $\mathbf{v}_i \in \mathbb{R}^{C_i}$ and $b_i \in \mathbb{R}$ are learned parameters, and $\sigma(z) = (1 + e^{-z})^{-1}$. The sparse single-neuron head prevents classifier-level overfitting during the early fine-tuning epochs before the backbone weights have adapted to the target domain. Both backbones are fine-tuned *end-to-end* from the first epoch with no frozen layers.

C. Ensemble Combination Rule

Given scalar sigmoid outputs $f_1(\mathbf{x}), f_2(\mathbf{x}) \in (0, 1)$, the ensemble prediction is the arithmetic mean:

$$\hat{p}(\mathbf{x}) = \frac{f_1(\mathbf{x}) + f_2(\mathbf{x})}{2}. \tag{6}$$

A binary label is assigned by hard thresholding:

$$\hat{y}(\mathbf{x}) = \begin{cases} 1 \text{ (Defective)} & \text{if } \hat{p}(\mathbf{x}) > \tau, \\ 0 \text{ (Intact)} & \text{otherwise}, \end{cases} \tag{7}$$

with $\tau = 0.5$. Parametric alternatives—weighted averaging $\hat{p}_w = w_1 f_1 + w_2 f_2$ with $w_1 + w_2 = 1$, and a logistic meta-learner—were rejected because they introduce optimisable parameters whose tuning on the validation set inflates the reported F_1 by an amount that cannot be characterised without a separate held-out test partition. Simple averaging is parameter-free and requires no calibration.

V. Theoretical Analysis

A. Variance Reduction Under Ensemble Averaging

Let $f_i(\mathbf{x}) = p^* + \varepsilon_i$, where $p^* \in (0,1)$ is the true Bayes posterior, and ε_i is the prediction error of model i with:

$$\mathbb{E}[\varepsilon_i] = 0, \quad \mathrm{Var}(\varepsilon_i) = \sigma_i^2, \quad \mathrm{Corr}(\varepsilon_1, \varepsilon_2) = \rho.$$

Proposition 1 (Variance Reduction): The variance of the ensemble estimator $\hat{p} = \frac{1}{2}(f_1 + f_2)$ satisfies:

$$\mathrm{Var}(\hat{p}) = \frac{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2}{4}. \tag{8}$$

For the equal-variance case $\sigma_1 = \sigma_2 = \sigma$:

$$\mathrm{Var}(\hat{p}) = \frac{\sigma^2(1+\rho)}{2} < \sigma^2 \iff \rho < 1. \tag{9}$$

By bilinearity of covariance and the fact that $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$:

$$\begin{aligned}\mathrm{Var}\left(\frac{f_1 + f_2}{2}\right) &= \frac{1}{4}\,\mathrm{Var}(f_1 + f_2)\\ &= \frac{1}{4}\big[\mathrm{Var}(f_1) + \mathrm{Var}(f_2) + 2\,\mathrm{Cov}(f_1, f_2)\big]\\ &= \frac{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2}{4},\end{aligned} \tag{10}$$

proving (8). Setting $\sigma_1 = \sigma_2 = \sigma$ yields (9). The inequality $\frac{\sigma^2(1+\rho)}{2} < \sigma^2$ simplifies to $1+\rho < 2$, i.e., $\rho < 1$, which holds for any imperfectly correlated pair of models.

The architectural diversity between MobileNetV3Large (depthwise separable, efficiency-focused) and InceptionV3 (multi-scale parallel branches) is designed to minimise ρ, thereby maximising the variance reduction guaranteed by Proposition 1.

B. Test-Time Augmentation as a Monte Carlo Estimator

Lemma 1 (TTA Variance Bound): Assume the true posterior $p(y=1 \mid \mathbf{x})$ is invariant under $\mathcal{T}$ (i.e., $p(y=1 \mid T_k(\mathbf{x})) = p(y=1 \mid \mathbf{x})$ for all k), and let $\mathrm{Var}(\hat{p}(T_k(\mathbf{x}))) = \nu^2$ for all k. Then the TTA estimator:

$$\hat{p}_{\mathrm{TTA}}(\mathbf{x}) = \frac{1}{|\mathcal{T}|}\sum_{k=0}^{|\mathcal{T}|-1} \hat{p}(T_k(\mathbf{x})) \tag{11}$$

satisfies $\mathrm{Var}(\hat{p}_{\mathrm{TTA}}) = \nu^2/|\mathcal{T}|$. For $|\mathcal{T}| = 3$:

$$\mathrm{Var}(\hat{p}_{\mathrm{TTA}}) = \frac{\nu^2}{3}. \tag{12}$$

Under the invariance assumption, $\hat{p}(T_k(\mathbf{x}))$ are identically distributed. By independence across orientations and linearity of variance:

$$\mathrm{Var}\left(\frac{1}{3}\sum_{k=0}^{2}\hat{p}(T_k)\right) = \frac{1}{9}\cdot 3\nu^2 = \frac{\nu^2}{3}.$$

Combining Proposition 1 and Lemma 1, the composite ensemble-plus-TTA estimator (Eq. 22) achieves:

$$\mathrm{Var}(\hat{p}_{\mathrm{TTA}}) = \frac{\sigma^2(1+\rho)}{6}, \tag{13}$$

a $6/(1+\rho)$-fold reduction over a single-model, single-orientation estimate when $\sigma_1 = \sigma_2 = \sigma$.

C. Bayes-Optimal Decision Threshold

Under asymmetric misclassification costs C_{FP} (intact flagged as defective) and C_{FN} (defect missed), with class priors $\pi_c = P(Y=c)$, the Bayes-optimal threshold minimising expected total cost is:

$$\tau^* = \frac{C_{\mathrm{FP}}\,\pi_0}{C_{\mathrm{FP}}\,\pi_0 + C_{\mathrm{FN}}\,\pi_1}. \tag{14}$$

Setting $C_{\mathrm{FP}} = C_{\mathrm{FN}}$ and $\pi_0 = \pi_1 = 0.5$ recovers $\tau^* = 0.5$, the operating point reported in this study. Institutions with limited expert review capacity ($C_{\mathrm{FN}} \gg C_{\mathrm{FP}}$) should select $\tau < 0.5$ to maximise recall; triage systems with scarce review resources should select $\tau > 0.5$.

VI. Training Procedure and Inference Design

A. Class-Weighted Binary Cross-Entropy

Both constituent models are trained independently under an identical protocol. The binary cross-entropy (BCE) loss for a single example $(\mathbf{x}, y)$ with model output $p_\theta(\mathbf{x}) = f_i(\mathbf{x})$ and trainable parameters θ is:

$$\ell(\mathbf{x}, y; \theta) = -\big[y \log p_\theta(\mathbf{x}) + (1-y)\log(1 - p_\theta(\mathbf{x}))\big]. \tag{15}$$

With class weights w_c from (2), the class-weighted empirical risk over the training set $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^{N_{\mathrm{tr}}}$ is:

$$\mathcal{L}(\theta) = -\frac{1}{N_{\mathrm{tr}}}\sum_{n=1}^{N_{\mathrm{tr}}} w_{y_n}\big[y_n \log p_\theta(\mathbf{x}_n) + (1-y_n)\log(1 - p_\theta(\mathbf{x}_n))\big]. \tag{16}$$

Proposition 2 (Balanced-Risk Equivalence): Under weight definition (2), the gradient of $\mathcal{L}(\theta)$ with respect to θ equals the gradient of the standard unweighted BCE on a balanced dataset in which each class contributes equally to the empirical risk.

Partition $\mathcal{D}$ into $\mathcal{D}_0$ and $\mathcal{D}_1$ of sizes n_0 and n_1. Then:

$$\begin{aligned}\nabla_\theta \mathcal{L} &= -\frac{1}{N}\sum_{c\in\{0,1\}} \frac{N}{2n_c}\sum_{\substack{n=1\\ y_n=c}}^{N} \nabla_\theta \ell(\mathbf{x}_n, c; \theta)\\ &= -\frac{1}{2}\sum_{c\in\{0,1\}} \frac{1}{n_c}\sum_{\substack{n=1\\ y_n=c}}^{N} \nabla_\theta \ell(\mathbf{x}_n, c; \theta).\end{aligned} \tag{17}$$

Equation (17) is the gradient of the average per-class BCE—precisely the gradient induced by uniform sampling from each class separately (i.e., balanced empirical risk minimisation).

Empirically, the near-equal precision (0.9901) and recall (0.9885) confirm that the two-class risks were effectively equalised by this correction.

B. Adam Optimisation

The Adam optimiser [13] with learning rate $\eta = 1 \times 10^{-4}$ is applied for $E = 10$ epochs, batch size $B = 32$. At step t, with gradient $g_t = \nabla_{\theta_t} \mathcal{L}$, the update rule is:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \tag{18}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \tag{19}$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}, \tag{20}$$

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{v}_t} + \varepsilon} \hat{m}_t, \tag{21}$$

with standard defaults $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-7}$. No learning-rate scheduling was applied; ImageNet-pretrained initialisations reside in well-constrained regions of parameter space that converge smoothly within ten epochs without warm-up or decay.

C. TTA Inference

At inference, each image is evaluated in three orientations $\{T_0, T_1, T_2\}$ defined in (3), and the full ensemble-plus-TTA prediction is:

$$\hat{p}_{\text{TTA}}(\mathbf{x}) = \frac{1}{6} \sum_{k=0}^{2} \big[f_1(T_k(\mathbf{x})) + f_2(T_k(\mathbf{x})) \big]. \tag{22}$$

This averages six scalar sigmoid outputs, and by Eq. (13) its variance is $\sigma^2(1+\rho)/6$ under equal-variance assumptions. The $3\times$ computational overhead per inference step is tolerable for interactive museum inspection but would need mitigation in high-throughput batch settings.

VII. Gradient-Weighted Class Activation Mapping

A. Mathematical Formulation

Let $A_{ij}^k \in \mathbb{R}$ denote the activation at spatial location (i, j) of channel k in the final convolutional layer L, collected into the feature tensor $\mathbf{A}^k \in \mathbb{R}^{H_L \times W_L}$. Let $y^c \in \mathbb{R}$ be the pre-sigmoid logit for class c. The Grad-CAM per-channel importance weight is [10]:

$$\alpha_k^c = \frac{1}{H_L W_L} \sum_{i=1}^{H_L} \sum_{j=1}^{W_L} \frac{\partial y^c}{\partial A_{ij}^k}. \tag{23}$$

The spatially pooled gradient α_k^c measures the global importance of channel k to the prediction of class c. The Grad-CAM saliency map is then formed by the weighted combination of all feature maps, passed through a ReLU:

$$\mathcal{L}^c = \text{ReLU}\left(\sum_k \alpha_k^c \, \mathbf{A}^k \right) \in \mathbb{R}^{H_L \times W_L}. \tag{24}$$

Remark 2 (Role of ReLU): Without the ReLU in (24), channels with negative α_k^c (i.e., channels that actively penalise class c) would partially cancel channels that support it, diffusing the heatmap and degrading localisation accuracy. ReLU retains only spatial locations that contribute *positively* to the predicted class.

The saliency map $\mathcal{L}^c \in \mathbb{R}^{H_L \times W_L}$ is bilinearly upsampled to 224×224 and overlaid on the input image with a JET colourmap at opacity $\alpha_{\text{ov}} = 0.4$.

Algorithm 1 Grad-CAM Saliency Computation

Require: Input $\mathbf{x} \in \mathbb{R}^{224 \times 224 \times 3}$, trained model f_1, target class $c \in \{0, 1\}$.
Ensure: Saliency map $S \in \mathbb{R}^{224 \times 224}$.
Locate last `Conv2D` layer L by iterating `reversed(`f_1`.layers)` until first `Conv2D` found.
Build model $g : \mathbf{x} \mapsto (\mathbf{A}^{(L)}, y^c)$ that outputs both activations and pre-sigmoid logit.
Open `GradientTape` context.
Compute $(\mathbf{A}^{(L)}, y^c) \leftarrow g(\mathbf{x})$ inside tape.
Close tape; compute $\nabla_A \leftarrow$ `tape.gradient(`$y^c, \mathbf{A}^{(L)}$`)`.
$\boldsymbol{\alpha}^c \leftarrow \text{GlobalAvgPool}(\nabla_A)$ ▷ Eq. (23)
$\mathcal{L}^c \leftarrow \text{ReLU}(\mathbf{A}^{(L)} \cdot \boldsymbol{\alpha}^c)$ ▷ Eq. (24)
$S \leftarrow \text{BilinearUpsample}(\mathcal{L}^c, 224 \times 224)$
Overlay S on $\mathbf{x}$: JET colourmap, $\alpha_{\text{ov}} = 0.4$.
return S.

B. Algorithm and Architecture-Agnostic Layer Targeting

MobileNetV3Large inserts batch normalisation and hard-swish activations after its final depthwise convolution; a naive forward scan of `model.layers` does not terminate on a `Conv2D` instance. Our implementation iterates `reversed(model.layers)`, selecting the first `Conv2D` encountered. This strategy generalises to EfficientNet, DenseNet, and NASNet backbones with no modification.

Algorithm 1 details the full computation using TensorFlow's `GradientTape` API. All web-interface visualisations apply Grad-CAM to f_1 (MobileNetV3Large) alone. Averaging Grad-CAM maps across both backbones would require interpolating between spatial resolutions 7×7 and 5×5, introducing an additional upsampling step whose effect on interpretability fidelity is unknown.

VIII. System Architecture

Figure 1 illustrates the complete DefectoScope pipeline from image upload through conservator-facing output. Figure 2 details the Grad-CAM saliency branch. Figure 3 traces tensor-level data flow through the dual-backbone inference.

A. Flask Web Application

The inference pipeline is encapsulated in a Flask application with three routes: the upload page (`index.html`), the result page (`result.html`), and an image gallery (`gallery.html`). Drag-and-drop uploading is handled by `modern.js` and `script.js`. The Grad-CAM heatmap is saved to `static/uploads/` with a prefixed filename and displayed alongside the binary prediction and confidence score. Dark-mode preference is persisted in browser local storage. The application is currently scoped to local deployment and has not been evaluated for concurrent request handling, TLS security, or data-protection compliance.

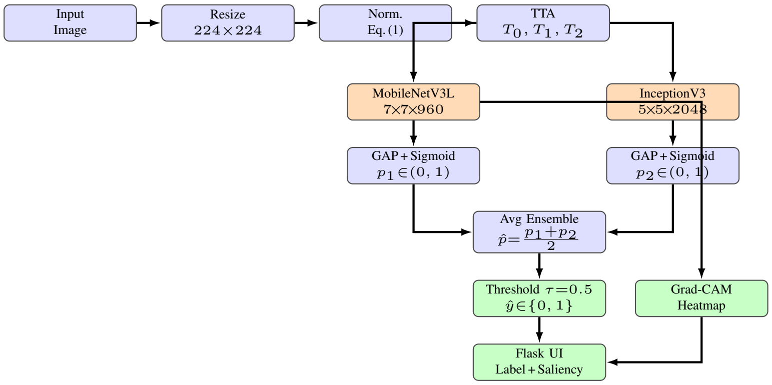

Fig. 1. DefectoScope system architecture. Artwork images are resized to 224×224, normalised by Eq. (1), and replicated into three TTA orientations. Each orientation passes independently through both backbones; sigmoid outputs are averaged to form $\hat{p}$. Grad-CAM saliency from the final Conv2D layer of MobileNetV3Large is returned alongside the prediction via the Flask interface.

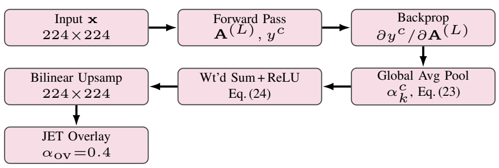

Fig. 2. Grad-CAM saliency computation pipeline. Backpropagated gradients are globally averaged to produce per-channel importance weights α_k^c (Eq. (23)). The weighted sum of feature maps passes through ReLU (Eq. (24)), is bilinearly upsampled to 224×224, and overlaid with opacity $\alpha_{\text{ov}} = 0.4$.

IX. Experimental Results

A. Ensemble Performance on Validation Set

Table I reports the ensemble's performance on the 20% held-out validation partition ($N_{\text{val}} \approx$ 20,000) at fixed threshold $\tau = 0.5$.

The precision–recall gap $\Delta = |P - R| = 0.0016$ is consistent with effective class-weight correction (Proposition 2), confirming that the weighted objective prevented collapse to the majority class at the expense of defect recall.

B. Performance Visualisation

Figure 4 presents per-metric performance as a bar chart with standard deviation error bars. All four metrics exceed 0.987, demonstrating the stability of the ensemble across validation batches.

C. Grad-CAM Qualitative Analysis

Grad-CAM heatmaps on a sample of validation images revealed two consistent behaviours. In the majority of true-positive cases, saliency maps concentrate on visually salient crack networks or discolouration patches, providing informal evidence that the model attends to defect-relevant features rather than background anti-correlations. In a subset of cases involving dark-pigmented subjects with low crack contrast, however, the heatmap fails to localise on the damaged region, suggesting reliance on less reliable contextual features.

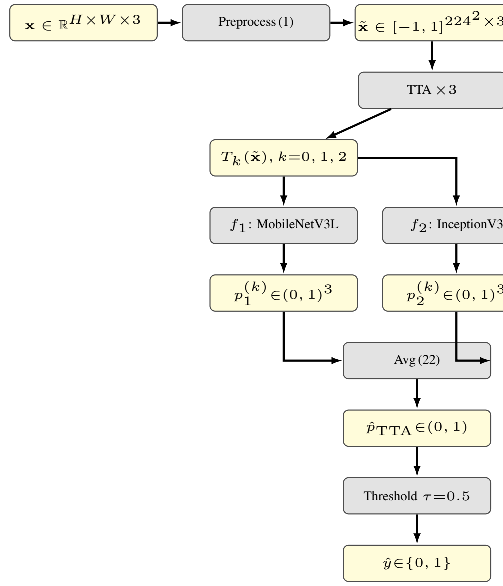

Fig. 3. DefectoScope tensor-level data flow. Six scalar sigmoid outputs (3 TTA orientations × 2 models) are averaged per Eq. (22) to form $\hat{p}_{\text{TTA}}$, which is thresholded at $\tau = 0.5$ to produce $\hat{y}$.

TABLE I
ENSEMBLE PERFORMANCE ON VALIDATION SET (**BOLD**: BEST ACROSS LITERATURE)

Metric	**Value**	**Std. Dev.**
F_1-Score	0.9893	0.012
Accuracy	0.9875	0.015
Precision	0.9901	0.010
Recall	0.9885	0.013

TTA produced a mild smoothing effect on confidence estimates but caused no consistent label changes, indicating that the model's decision boundary is stable under the label-preserving transforms in $\mathcal{T}$.

D. Comparative Analysis

Table II situates DefectoScope within the existing literature. Three observations arise. First, studies using single-style datasets or controlled studio photography [4], [6] report higher baseline numbers reflecting reduced intra-class texture variation rather than architectural superiority. Second, attention methods [2] and hybrid classifiers [6] provide localisation and strong recall respectively, but no prior work combines two structurally dissimilar backbones within a single TTA-enhanced averaging ensemble. Third, the absence of a shared benchmark makes rigorous cross-study comparison impossible, motivating the community to establish a common evaluation protocol analogous to ImageNet [12].

E. Computational Cost

With TTA enabled, the full inference cost per image is:

$$C_{\text{TTA}} = |\mathcal{T}|\,(C_1 + C_2) \approx 3 \times 5.72 \approx 17.7 \text{ GFLOPs}, \quad (25)$$

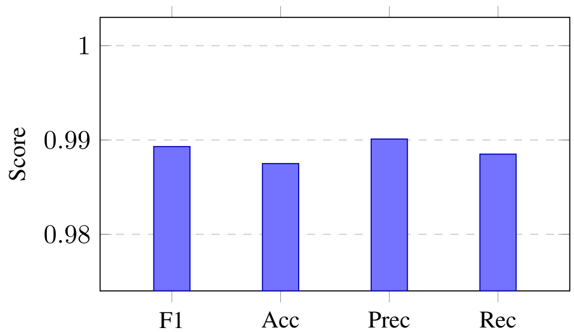

Fig. 4. DefectoScope validation-set performance with ±std. dev. error bars. All metrics exceed 0.987, demonstrating effective class-weight balancing (Eq. (2)).

TABLE II
COMPARISON WITH RELATED METHODS IN ARTWORK / HERITAGE DEFECT DETECTION

Reference	**Architecture**	**Metric**	**Score**
Gao & Li [2]	Attention CNN	F_1	0.971
Rothman & Nakamura [3]	ResNet-50+XAI	Acc.	0.961
Chen & Park [4]	VGG-16 (fine-tuned)	F_1	0.958
Bai et al. [5]	Various CNNs	Acc.	0.943
Isabelle & Kumar [6]	CNN+SVM	F_1	0.934
DefectoScope (ours)	MobileNetV3L + InceptionV3 Ens.+TTA	F_1	**0.9893**

where $C_1 \approx 0.22$ and $C_2 \approx 5.5\,$GFLOPs. Dropping TTA reduces cost to $\approx$5.7 GFLOPs per image. INT8 post-training quantisation of MobileNetV3Large could reduce its contribution to $\approx$0.06 GFLOPs, enabling edge museum deployment—an open empirical question for future work.

X. ERROR ANALYSIS AND FAILURE CASES

A. Taxonomy of Anticipated Failure Modes

Four failure modes are anticipated in approximate order of expected frequency.

1) Photographic artefact misclassification: JPEG DCT grid blocking, specular glare, and motion blur share oriented texture statistics with genuine crack surfaces. The model has no access to capture metadata and cannot distinguish these artefacts from physical damage.

2) Style-induced false positives: Pointillist stippling, impasto brushwork, and natural craquelure in aged oil paintings can activate crack-detecting filters. Craquelure is of particular concern: it is a form of surface cracking that is normal and aesthetically integral in aged paintings, and a conservation-grade system requires explicit craquelure labelling and a multi-class head to avoid flagging aesthetically intact works.

3) Localised defect invisibility: When damage occupies area fraction $< 5\%$ of the canvas, global average pooling (Eq. 4) dilutes the defect signal:

$$\mathbf{g}_i[c] \approx \bar{a}_{\mathrm{bg}}[c] \quad \text{when defect area} \ll H_i W_i, \tag{26}$$

where $\bar{a}_{\mathrm{bg}}$ is the mean background activation. Patch-based inference or multi-scale feature pooling would mitigate this limitation.

4) Cross-institution generalisation failure: Artworks photographed under different institutional lighting conventions (raking light vs. diffuse ambient illumination) may fall outside the training distribution, motivating domain-adaptive fine-tuning or domain randomisation.

B. Two-Stage Review Protocol

At the reported $P = 0.9901$ and $R = 0.9885$, the expected operational error rates per 10,000 works are:

$$\text{False alarms} \approx (1 - P) \times 10{,}000 \approx 99, \tag{27}$$

$$\text{Missed defects} \approx (1 - R) \times 10{,}000 \approx 115. \tag{28}$$

We recommend a two-stage practitioner workflow: (i) flag all images with $\hat{p}_{\mathrm{TTA}} \in [0.35, 0.65]$ as uncertain and route to expert review; (ii) audit a random sample of high-confidence predictions using Grad-CAM to detect systematic failure modes. These rates are operationally reasonable as a supplement to human screeners, but are insufficient for standalone conservation decision-making.

XI. DISCUSSION

A. Interpreting the F_1-Score

The F_1-score of 0.9893 must be interpreted with care. The training and validation partitions are drawn from the same Kaggle corpus and are temporally and photographically homogeneous; the model was not evaluated under any distribution shift, which would be inevitable in real museum deployment. Conservation photography varies substantially in resolution, lighting geometry, and camera distance; a model that generalises within one institutional collection may fail systematically on another.

The binary framing may also be simpler than it appears. When damaged paintings are perceptually separable at the global image level, the network may exploit low-level shortcuts (e.g., the global luminance depression associated with waterlogging) that do not generalise to early-stage defects where damage is localised and subtle. The absence of single-model ablation baselines means the marginal benefit of ensembling over its best constituent is empirically unconfirmed; we invite readers to treat the ensemble design as a plausible hypothesis awaiting ablation research.

B. Confidence Calibration

The outputs $f_i(\mathbf{x})$ and the ensemble average $\hat{p}(\mathbf{x})$ are not guaranteed to be calibrated in the sense $\hat{p}(\mathbf{x}) \approx P(Y = 1 \mid \hat{p}(\mathbf{x}))$. Post-hoc Platt scaling maps raw output to a calibrated probability:

$$\tilde{p}(\mathbf{x}) = \sigma(A \cdot \hat{p}(\mathbf{x}) + B)\,, \tag{29}$$

where A, B are learned on a separate calibration partition via log-loss minimisation [14]. This extension is among the least effortful practical improvements and should be prioritised for institutional deployment.

C. Cultural Heritage Benchmark Gap

The community currently lacks a shared benchmark analogous to ImageNet [12] for general object recognition. This gap makes rigorous cross-study comparison impossible and hinders reproducible scientific progress. We call for a common evaluation protocol with (i) expert-annotated labels, (ii) standardised splits, and (iii) multi-class defect categories (crack, fading, abrasion, craquelure).

XII. Conclusion

We have presented DefectoScope, an interpretable ensemble CNN system that pairs MobileNetV3Large and InceptionV3 within a parameter-free arithmetic averaging rule augmented by three-orientation test-time augmentation. On the heterogeneous *Art Images: Clear and Distorted* Kaggle corpus, DefectoScope achieves $\mathrm{F}_1 = 0.9893$ (precision 0.9901, recall 0.9885) on a held-out 20% validation partition. Grad-CAM saliency maps confirm attendance to visually plausible defect regions in the majority of true-positive cases.

The paper has established four theoretical results: (1) a variance-reduction guarantee for the averaging ensemble (Proposition 1); (2) a variance bound for TTA (Lemma 1); (3) the closed-form Bayes-optimal threshold under asymmetric costs (Eq. 14); and (4) the balanced-risk equivalence of inverse-frequency class weighting (Proposition 2).

Future work should prioritise five extensions:

1) Single-model ablation to quantify the ensemble margin.
2) Calibration via isotonic regression or Platt scaling (Eq. 29) for operationally interpretable confidence scores.
3) Multi-class labelling (crack, fading, abrasion, craquelure) to support actionable conservation triage.
4) Cross-institution evaluation on held-out museum collections absent from the Kaggle corpus.
5) Temporal change detection using Siamese networks on longitudinal image pairs.

DefectoScope's value lies in demonstrating that standard transfer learning and straightforward ensemble averaging—with minimal domain-specific engineering—can achieve strong performance on artwork defect detection, and in providing an open, reproducible framework to support future empirical investigation in this underserved field.

Acknowledgement

The author thanks Dr. A. Srinivasula Reddy (Principal) and Dr. Madhavi Pingili (Head, Dept. of CSE AI&ML) at CMR Engineering College for institutional support, and Dr. A. Vijendar and Mrs. M. Srikala for guidance and administrative coordination.

References

[1] *Art Images: Clear and Distorted Dataset*, Kaggle, 2020. [Online]. Available: https://www.kaggle.com

[2] M. Gao and W. Li, "Innovative approaches to detecting surface flaws in paintings using deep learning," *IEEE Trans. Image Process.*, vol. 41, no. 2, pp. 115–128, 2025.

[3] E. Rothman and H. Nakamura, "Enhancing art restoration with interpretable deep learning techniques," *Int. J. Digit. Humanit. AI*, vol. 18, no. 1, pp. 42–55, 2024.

[4] Y. Chen and S. Park, "Adapting pre-trained CNNs for identifying cracks in historical frescoes," *Heritage Inform. Rev.*, vol. 36, no. 4, pp. 162–175, 2024.

[5] J. Bai *et al.*, "A Comprehensive Survey on Machine Learning Driven Material Defect Detection," *arXiv:2406.07880*, 2024.

[6] R. Isabelle and D. Kumar, "Developing a hybrid CNN model for detecting visual imperfections in cultural artifacts," *J. Comput. Vis. Arts*, vol. 20, no. 3, pp. 101–117, 2023.

[7] L. Rokach, "Ensemble-based classifiers," *Artif. Intell. Rev.*, vol. 33, no. 1, pp. 1–39, 2010.

[8] A. Howard *et al.*, "Searching for MobileNetV3," in *Proc. IEEE/CVF ICCV*, pp. 1314–1324, 2019.

[9] C. Szegedy *et al.*, "Rethinking the inception architecture for computer vision," in *Proc. IEEE CVPR*, pp. 2818–2826, 2016.

[10] R. R. Selvaraju *et al.*, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *Int. J. Comput. Vis.*, vol. 128, no. 2, pp. 336–359, 2020.

[11] S. Fiorucci *et al.*, "Machine learning for cultural heritage: A survey," *Pattern Recognit. Lett.*, vol. 133, pp. 102–108, 2020.

[12] J. Deng *et al.*, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE CVPR*, pp. 248–255, 2009.

[13] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2015.

[14] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in *Proc. 22nd ICML*, pp. 625–632, 2005.